Name : Sanjana

SRN : PES1UG24CS420

1 A)

```
sanjana@DESKTOP-SAN:~/PES1UG24CS420-pes-vcs$ make test_objects
gcc -Wall -Wextra -O2 -c test_objects.c -o test_objects.o
gcc -Wall -Wextra -O2 -c object.c -o object.o
gcc -o test_objects test_objects.o object.o -lcrypto
sanjana@DESKTOP-SAN:~/PES1UG24CS420-pes-vcs$ ./test_objects
Stored blob with hash: d58213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
Object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
PASS: blob storage
PASS: deduplication
PASS: integrity check

All Phase 1 tests passed.
sanjana@DESKTOP-SAN:~/PES1UG24CS420-pes-vcs$
```

1 B)

```
sanjana@DESKTOP-SAN:~/PES1UG24CS420-pes-vcs$ make pes
gcc -Wall -Wextra -O2 -c tree.c -o tree.o
gcc -Wall -Wextra -O2 -c index.c -o index.o
gcc -Wall -Wextra -O2 -c commit.c -o commit.o
gcc -Wall -Wextra -O2 -c pes.c -o pes.o
gcc -o pes object.o tree.o index.o commit.o pes.o -lcrypto
sanjana@DESKTOP-SAN:~/PES1UG24CS420-pes-vcs$ ./pes init
Initialized empty PES repository in .pes/
sanjana@DESKTOP-SAN:~/PES1UG24CS420-pes-vcs$ find .pes/objects -type f
.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476
sanjana@DESKTOP-SAN:~/PES1UG24CS420-pes-vcs$
```

2 A

```
sanjana@DESKTOP-SAN:~/PES1UG24CS420-pes-vcs$ gcc -Wall -Wextra -O2 -c index.c -o index.o
sanjana@DESKTOP-SAN:~/PES1UG24CS420-pes-vcs$ gcc -o test_tree test_tree.o object.o tree.o index.o -lcrypto
sanjana@DESKTOP-SAN:~/PES1UG24CS420-pes-vcs$ ./test_tree
Serialized tree: 139 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization

All Phase 2 tests passed.
```

2 B

```
sanjana@DESKTOP-SAN:~/PES1UG24CS420-pes-vcs$ xxd .pes/objects/2c/f8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4 | head -20
00000000: 626c 6f62 2036 0068 656c 6c6f 0a         blob 6.hello.
sanjana@DESKTOP-SAN:~/PES1UG24CS420-pes-vcs$
```

3 A

```
sanjana@DESKTOP-SAN:~/PES1UG24CS420-pes-vcs$ cat .pes/index
100644 2cf8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4 1776673101 6 file1.txt
100644 e00c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f 1776673107 6 file2.txt
sanjana@DESKTOP-SAN:~/PES1UG24CS420-pes-vcs$
```

3 B

```
sanjana@DESKTOP-SAN:~/PES1UG24CS420-pes-vcs$ find .pes/objects -type f
.pes/objects/2c/f8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4
.pes/objects/e0/0c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f
sanjana@DESKTOP-SAN:~/PES1UG24CS420-pes-vcs$
```

4 A

```
sanjana@DESKTOP-SAN:~/PES1UG24CS420-pes-vcs$ find .pes -type f | sort
.pes/HEAD
.pes/index
.pes/objects/2c/f8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4
.pes/objects/34/fb83796d921969e6ba2a0c30dac825ea371b12576ff5709ffb9064f2112da4
.pes/objects/3a/0e105a52a08827a38204aa8d148e081bbca6a55fe29c0df6acfb58d84473c0
.pes/objects/50/9adf088abf8f298e70b59ce0576b9b29d710d9e1619788e53044ff5bb265b4
.pes/objects/64/b09382bfb8109ddf3bb83478bd047d54c269ad55143b9136d0bd315d10ff34
.pes/objects/b4/e76aee38854839557e5a3132e346adbf8288ec2ba878f2cd915d5c2326af00
.pes/objects/ce/ed54744634f312e9a7da9be821ec651ce533d1629bf9dd90608d325e896893
.pes/objects/e0/0c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f
.pes/objects/e6/7ed66bcb4a708250a89f315d4d8bb92703343c84df834438880e13a68652bc
.pes/objects/f5/902ee73b6aa25436f687067fc4cc0cb481cf416740d3c3c3ed6d9c4de7b6c2
.pes/refs/heads/main
sanjana@DESKTOP-SAN:~/PES1UG24CS420-pes-vcs$
```

4 B

```
sanjana@DESKTOP-SAN:~/PES1UG24CS420-pes-vcs$ cat .pes/refs/heads/main
3a0e105a52a08827a38204aa8d148e081bbca6a55fe29c0df6acfb58d84473c0
sanjana@DESKTOP-SAN:~/PES1UG24CS420-pes-vcs$ cat .pes/HEAD
ref: refs/heads/main
sanjana@DESKTOP-SAN:~/PES1UG24CS420-pes-vcs$
```

4 C

```
sanjana@DESKTOP-SAN:~/PES1UG24CS420-pes-vcs$ make test-integration
=== Running integration tests ===
bash test_sequence.sh
=== PES-VCS Integration Test ===

--- Repository Initialization ---
Initialized empty PES repository in .pes/
PASS: .pes/objects exists
PASS: .pes/refs/heads exists
PASS: .pes/HEAD exists

--- Staging Files ---
Status after add:
Staged changes:
  staged:     file.txt
  staged:     hello.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  (nothing to show)


--- First Commit ---
Committed: 90488c35f372... Initial commit

Log after first commit:
commit 90488c35f37253c8a529c18cd266e05728ad162d553c2cecd2bc076493b8dbfb
Author: Sanjana <sanjana@pes.edu>
Date:   1776677290

    Initial committial comQ


--- Second Commit ---
Committed: 163954fbd2c5... Update file.txt

--- Third Commit ---
Committed: e028ba547252... Add farewell

--- Full History ---
commit e028ba54725273c9cbe93c0053e3496863706e0cdfe08ec75f1a7995cad1f925
Author: Sanjana <sanjana@pes.edu>
Date:   1776677290

    Add farewelldd

commit 163954fbd2c5cb27ef143666b86c292482ce24b503614e4596e3d6265c06efab
Author: Sanjana <sanjana@pes.edu>
Date:   1776677290

    Update file.txt

commit 90488c35f37253c8a529c18cd266e05728ad162d553c2cecd2bc076493b8dbfb
Author: Sanjana <sanjana@pes.edu>
Date:   1776677290

    Initial commit
```

```
--- Reference Chain ---
HEAD:
ref: refs/heads/main
refs/heads/main:
e028ba54725273c9cbe93c0053e3496863706e0cdfe08ec75f1a7995cad1f925

--- Object Store ---
Objects created:
10
.pes/objects/0b/d69098bd9b9cc5934a610ab65da429b525361147faa7b5b922919e9a23143d
.pes/objects/10/1f28a12274233d119fd3c8d7c7216054ddb5605f3bae21c6fb6ee3c4c7cbfa
.pes/objects/16/3954fbd2c5cb27ef143666b86c292482ce24b503614e4596e3d6265c06efab
.pes/objects/58/a67ed1c161a4e89a110968310fe31e39920ef68d4c7c7e0d6695797533f50d
.pes/objects/90/488c35f37253c8a529c18cd266e05728ad162d553c2cecd2bc076493b8dbfb
.pes/objects/ab/5824a9ec1ef505b5480b3e37cd50d9c80be55012cabe0ca572dbf959788299
.pes/objects/b1/7b838c5951aa88c09635c5895ef7e08f7fa1974d901ce282f30e08de0ccd92
.pes/objects/d0/7733c25d2d137b7574be8c5542b562bf48bafeaa3829f61f75b8d10d5350f9
.pes/objects/db/07a1451ca9544dbf66d769b505377d765efae7adc6b97b75cc9d2b3b3da6ff
.pes/objects/e0/28ba54725273c9cbe93c0053e3496863706e0cdfe08ec75f1a7995cad1f925

=== All integration tests completed ===
sanjana@DESKTOP-SAN:~/PES1UG24CS420-pes-vcs$
```

Questions and Answers

Q5.1:
`pes checkout <branch>` updates HEAD to the target branch and loads its latest commit into the working directory. The index is also updated to match that commit snapshot. Files change because each branch has a different commit tree. The system must handle modified/untracked files carefully. Checkout may block if changes would be overwritten. This makes it complex due to safe state transitions.

Q5.2:
A dirty working directory is detected by comparing working directory, index, and HEAD. Any mismatch indicates uncommitted changes. This includes modified, deleted, or untracked files. Before checkout, the system checks if these changes will be overwritten. If yes, it blocks the operation. This prevents accidental loss of work.

Q5.3:
Detached HEAD means HEAD points to a commit, not a branch. Commits made here are not linked to any branch. These commits can be lost when switching branches. The history becomes dangling if not saved. Recovery is done by creating a branch at that commit. Reflog can also help restore lost commits.

Q6.1:
Start from all references like branches, tags, and HEAD. Traverse the commit graph using DFS or BFS. Mark all reachable objects (commits, trees, blobs). Unmarked objects are unreachable. Delete them to free space. For 100k commits, ~100k objects are visited due to shared history.

Q6.2:
GC may run while a commit is being created. New objects may appear unreachable during scanning. This can cause accidental deletion. Git avoids this using locks and coordination. It also delays deletion using a grace period. This ensures new commits are safely referenced.